

TP1 — Stateful Web Tracking

Web Privacy — UM6P

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1. Environment Setup

Virtual environment created, activated, and Flask installed successfully

```
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\OMAR>mkdir statefulTracking
A subdirectory or file statefulTracking already exists.

C:\Users\OMAR>cd statefulTracking

C:\Users\OMAR\statefulTracking>python -m venv venv

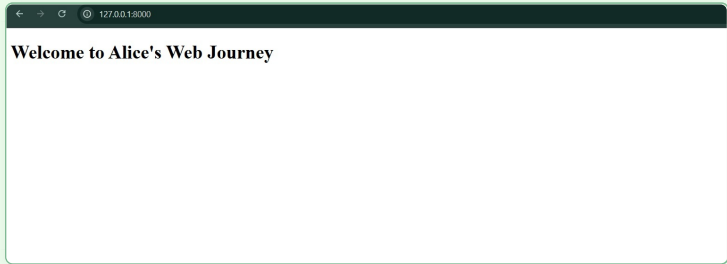
C:\Users\OMAR\statefulTracking>
C:\Users\OMAR\statefulTracking>venv\Scripts\activate

(venv) C:\Users\OMAR\statefulTracking>pip install flask
Collecting flask
  Downloading flask-3.1.3-py3-none-any.whl.metadata (3.2 kB)
Collecting blinker>=1.9.0 (from flask)
  Downloading blinker-1.9.0-py3-none-any.whl.metadata (1.6 kB)
Collecting click>=8.1.3 (from flask)
  Downloading click-8.5.0-py3-none-any.whl.metadata (2.6 kB)
Collecting itsdangerous>=2.2.0 (from flask)
  Downloading itsdangerous-2.2.0-py3-none-any.whl.metadata (1.9 kB)
Collecting jinja2>=3.1.2 (from flask)
  Using cached jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Collecting markupsafe>=2.1.1 (from flask)
  Using cached markupsafe-3.0.3-cp313-cp313-win_amd64.whl.metadata (2.8 kB)
Collecting werkzeug>=3.1.0 (from flask)
  Downloading werkzeug-3.1.8-py3-none-any.whl.metadata (4.0 kB)
Downloading flask-3.1.3-py3-none-any.whl (103 kB)
Downloading blinker-1.9.0-py3-none-any.whl (8.5 kB)
Downloading click-8.5.0-py3-none-any.whl (125 kB)
Downloading itsdangerous-2.2.0-py3-none-any.whl (16 kB)
Using cached jinja2-3.1.6-py3-none-any.whl (134 kB)
Using cached markupsafe-3.0.3-cp313-cp313-win_amd64.whl (15 kB)
Downloading werkzeug-3.1.8-py3-none-any.whl (226 kB)
Installing collected packages: markupsafe, itsdangerous, click, blinker, werkzeug, jinja2, flask
Successfully installed blinker-1.9.0 click-8.5.0 flask-3.1.3 itsdangerous-2.2.0 jinja2-3.1.6 markupsafe-3.0.3 werkzeug-3.1.8

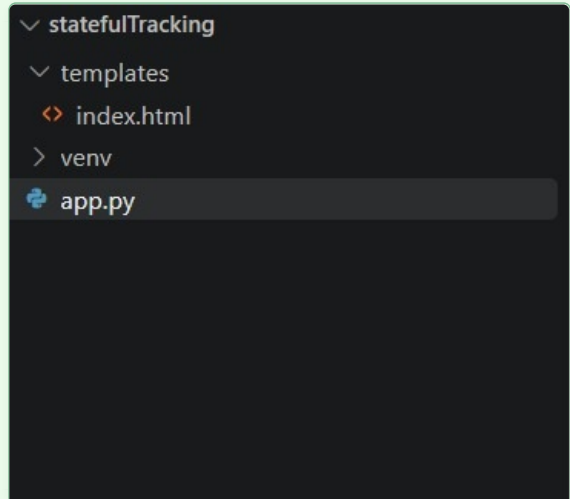
(venv) C:\Users\OMAR\statefulTracking>flask --version
Python 3.13.15
Flask 3.1.3
```

2. First Website

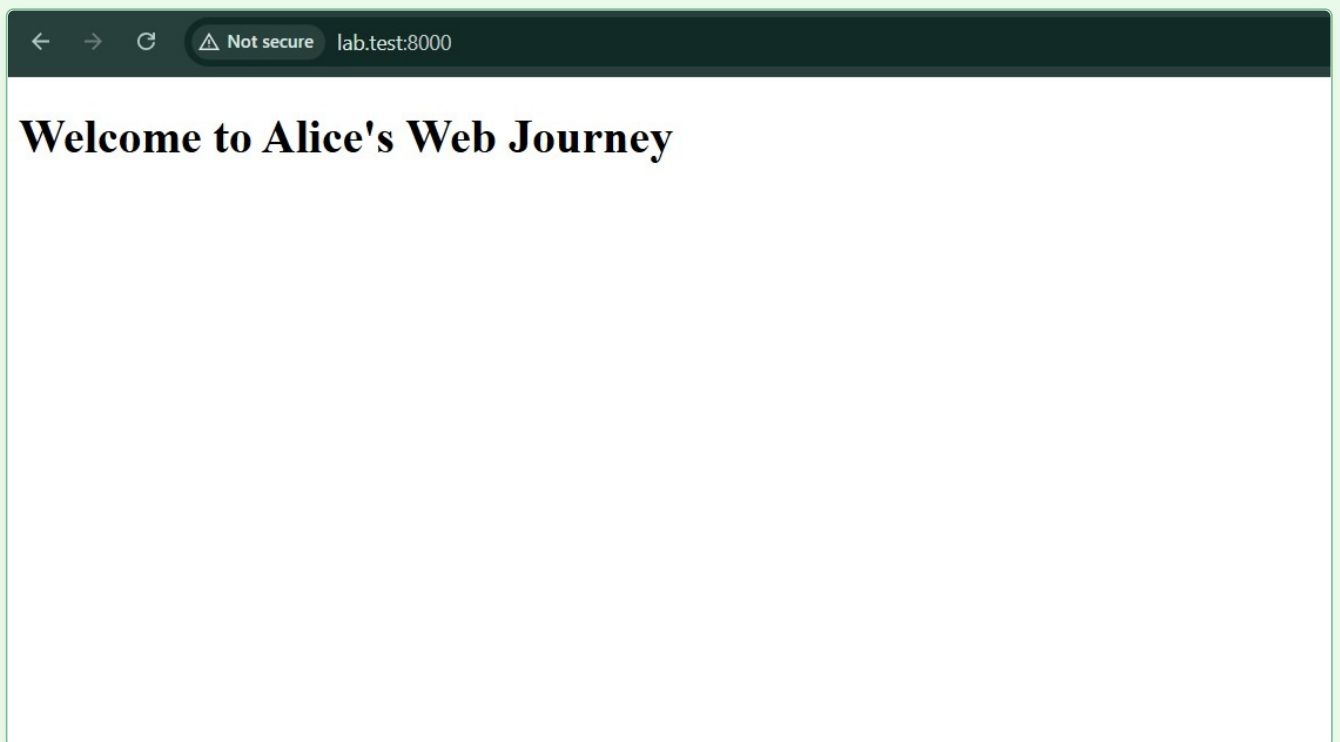
Base Flask server running — page accessible at
127.0.0.1:8000



Project structure: app.py and templates/index.html inside
statefulTracking



Site reached through the custom domain lab.test:8000 after editing the hosts file



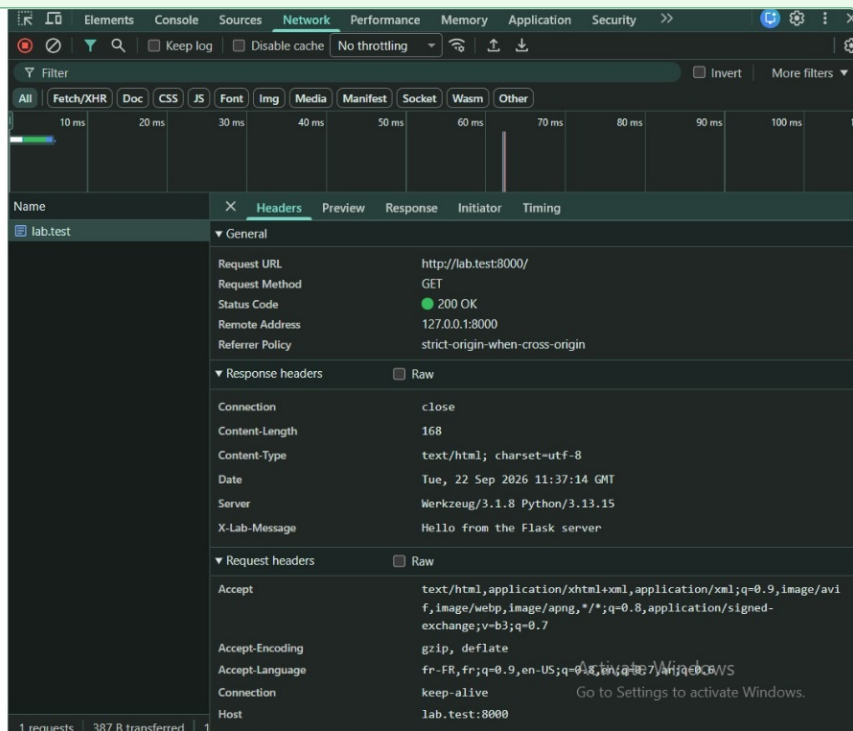
3. HTTP Headers

app.py updated to add a custom X-Lab-Message response header and print request/response headers server-side

```
Welcome X app.py X index.html X
app.py > ...
1 from flask import Flask, render_template, request, make_response
2
3 app = Flask(__name__)
4
5 @app.route("/")
6 def home():
7     response = make_response(render_template("index.html"))
8     response.headers["X-Lab-Message"] = "Hello from the Flask server"
9
10    print("\n--- HTTP REQUEST HEADERS ---")
11    print(request.headers)
12    print("--- HTTP RESPONSE HEADERS ---")
13    print(response.headers)
14
15    return response
16
17 if __name__ == "__main__":
18     app.run(host="127.0.0.1", port=8000, debug=True)
```

DevTools Network tab: the custom X-Lab-Message header is visible among the response headers

Welcome to Alice's Web Journey

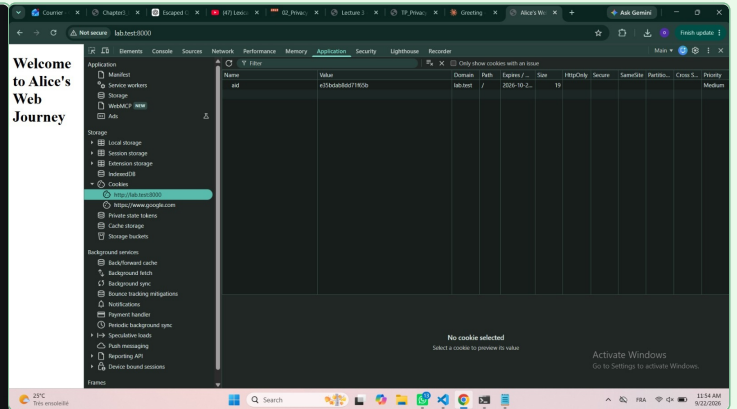


4. Cookie Manipulation

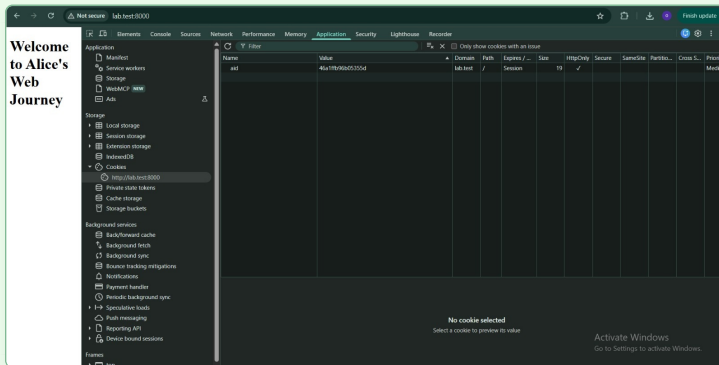
app.py updated to generate and set a persistent "aid" identifier cookie

```
1 from flask import Flask, render_template, request, make_response
2 import secrets
3
4 app = Flask(__name__)
5
6 @app.route("/")
7 def home():
8     aid = request.cookies.get("aid")
9     is_new = aid is None
10    if is_new:
11        aid = secrets.token_hex(8)
12
13    response = make_response(render_template("index.html"))
14
15    if is_new:
16        response.set_cookie(key="aid", value=aid)
17
18    return response
19
20 if __name__ == "__main__":
21     app.run(host="127.0.0.1", port=8000, debug=True)
```

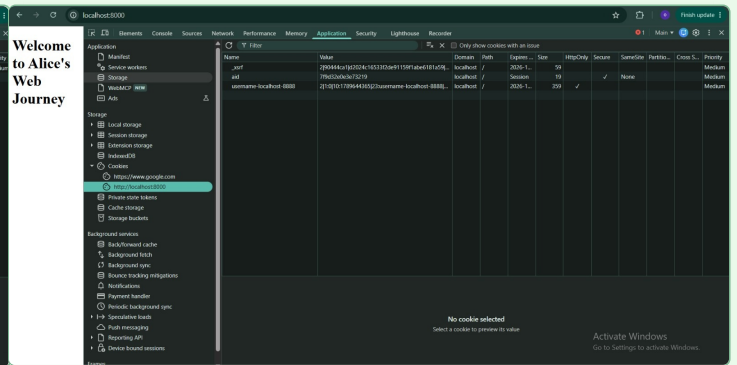
max_age: Expires changes from "Session" to a fixed date (~30 days later) — survives a browser restart



httponly=True: HttpOnly column is now checked — JavaScript can no longer read this cookie



secure=True, samesite="None": both flags set — cookie now restricted to HTTPS contexts



samesite="Strict": cookie marked Strict — will not be sent on cross-site requests

